

Nahid Hasan Sagar

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PROFESSIONAL SUMMARY

M.S. candidate in Statistical Science at Indiana University Bloomington with expertise in Bayesian modeling, statistical learning, and scientific computing, targeting industry statistician roles. Develops and benchmarks scalable research software in R, C++, and Python, with hands-on experience accelerating computational pipelines to optimize performance on fMRI data and applying Bayesian tree-based methods to financial data.

EDUCATION

Indiana University, College of Arts and Sciences

Master of Science in Statistical Science

Thesis: Bayesian Additive Regression Trees (BART) for Financial Data Analysis

Graduate Coursework: Introduction to Statistical Computing; Applied Statistical Computing; Fundamentals of Statistics I–II; Applied Linear Models I–II

Current Coursework: Statistical Consulting; Research in Statistics; Introduction to Analysis I

University of Dhaka

Bachelor of Science in Physics

Bloomington, IN

Expected May 2027

Dhaka, Bangladesh

August 2024

EXPERIENCE

Research Assistant, StatMIND Lab

Indiana University Bloomington

2025 – Present

Bloomington, IN

- Developed core functionality of Bayesian brain mapping algorithms developed by Dr. Amanda F. Mejia with the implementation of efficient C++ code with Rcpp and RcppArmadillo for high-dimensional neuroimaging data.
- Reproduced the exact results of the original R package implementation and validated the correctness of the new C++ code through extensive unit testing and benchmarking.
- Achieved a 4 to 5× speedup in core functional computations with proper implementation of C++ code inside R packages, enabling faster analysis of large-scale neuroimaging datasets.
- Build reproducible benchmarking workflows to compare runtime, scalability, and computational performance across implementations.

Teaching Assistant, Department of Statistics

Indiana University Bloomington

2025 – Present

Bloomington, IN

- Support undergraduate Statistics courses through classroom instruction, discussion sections, and individualized guidance.
- Evaluate approximately 200 assignments per grading cycle and provide detailed feedback on statistical reasoning, computation, and interpretation.
- Conduct weekly office hours and exam-review sessions to strengthen students' conceptual understanding and problem-solving skills.

Senior Mathematics Instructor

Udvash Academic & Admission Care

2016 – 2025

Dhaka, Bangladesh

- Taught and mentored more than 100,000 students through classroom instruction and national online learning platforms.
- Prepared students for competitive university admission examinations through structured lessons, practice materials, and academic planning.
- Translated advanced mathematical concepts into accessible problem-solving strategies for students with varied educational backgrounds.

SELECTED RESEARCH & PROJECTS

M.S. Thesis: Bayesian Additive Regression Trees for Financial Data

R, Bayesian Modeling

- Conduct thesis research on the use of Bayesian Additive Regression Trees for analyzing financial data under the supervision of Dr. Matthew T. Pratola.

TECHNICAL SKILLS

Statistical Methods: Bayesian inference, BART, mixed-effects models, bootstrap methods, experimental design, model selection

Statistical Learning: Ridge regression, lasso, decision trees, random forests, boosting, cross-validation, classification metrics, and predictive model evaluation

Programming: R, Python, C, C++, MATLAB, Bash, HTML, CSS, and JavaScript

Computing and Tools: Numerical optimization, algorithm benchmarking, code profiling, unit testing, Git, GitHub, LaTeX, RStudio, and VS Code

LEADERSHIP & HONORS

- Attended the 2026 Velocity Conference at UC Berkeley as an Indiana University representative for the science community.
- Collaborated on the development of a comprehensive online Linear Algebra curriculum during the COVID-19 pandemic.
- Co-organized the Australian Mathematics Competition in Bangladesh in 2014.